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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	LEX-Based Token Generator for a Simple Programming Language	
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Abstract

The project **"LEX-Based Token Generator for a Simple Programming Language"** focuses on the **lexical analysis phase of a compiler**, which is the first major phase of the compilation process. Lexical analysis is responsible for reading the source program and dividing it into meaningful units called **tokens**. The proposed system takes source code as input and scans it character by character to identify different types of tokens such as keywords, identifiers, numbers, operators, relational operators, and special symbols.

LEX/FLEX is used to develop the lexical analyzer by defining token patterns using **regular expressions**. Based on these predefined patterns, the system automatically recognizes the lexemes present in the source code and assigns the appropriate token type to each lexeme. The generated tokens are displayed in a structured token table, making the tokenization process easier to understand and analyze.

The system can also identify and report **invalid or unknown tokens**, helping users understand lexical errors in the input program. Additional information such as token numbers, lexemes, token types, and line numbers can be displayed to provide a clear representation of the lexical analysis process. The system can also provide token counts for different categories, such as keywords, identifiers, operators, numbers, and special symbols.

The main objective of this project is to provide a **simple, practical, and educational implementation of lexical analysis** using LEX/FLEX. It helps students understand how a compiler reads source code, recognizes patterns, and converts the input program into a sequence of tokens before further compilation phases such as syntax and semantic analysis. Thus, the project demonstrates the working of a lexical analyzer in an easy-to-understand and organized manner.

Signature of the Guide HOD